

UofT SURP - Introductory Assignment

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Part 1

- (A) Plotting the population SCYA69 in RA vs Dec sky coordinates and RA/Dec proper motion:

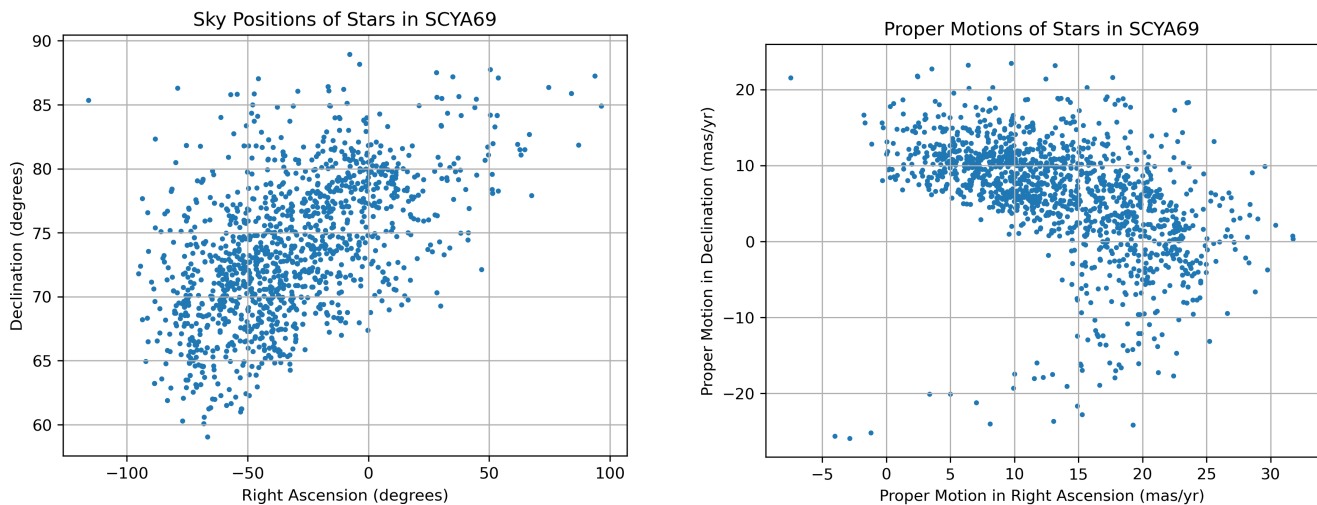


Figure 1: SCYA69 plotted in RA–Dec sky coordinates and RA/Dec proper motion space.

- (B) Converted the RA/Dec coordinates of SCYA69 to l/b galactic coordinates

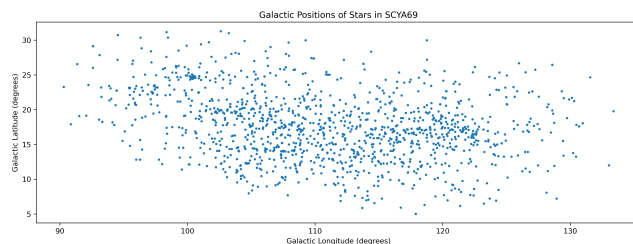


Figure 2: SCYA69 plotted in l/b galactic coordinates

- (C) Converting the RA, Dec, and distance of SCYA69 to galactic XYZ coordinates and plotting X vs Y, X vs Z, and Y vs Z.

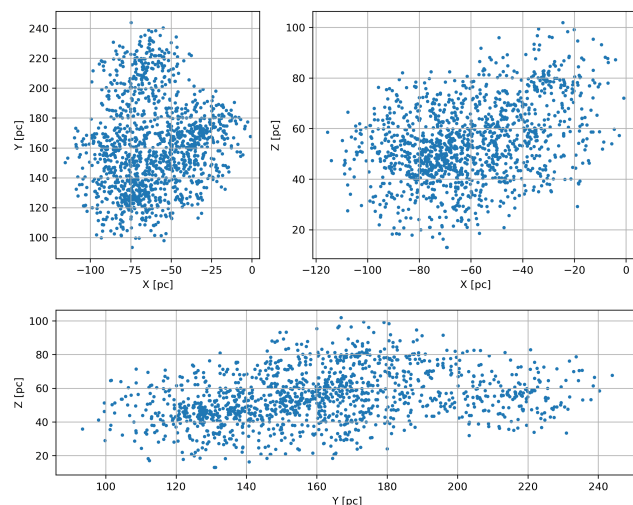


Figure 3: SCYA69 plotted in X vs Y, X vs Z, and Y vs Z galactic coordinates

- (D) Used the V_t function to convert proper motion and distance to transverse velocity (see notebook for conversion code)
- (E) Creating a color-magnitude diagram of CFN (SCYA 69). Plotting Gaia absolute G magnitude against Gaia bp-rp colour. The first plot plots this in three dimensions adding a colour map of P_{mem} to see where the highest probability young cluster members exist. The second plot restricts plot to members with $P_{mem} > 0.6$ to better reveal just the pre-main sequence branch.

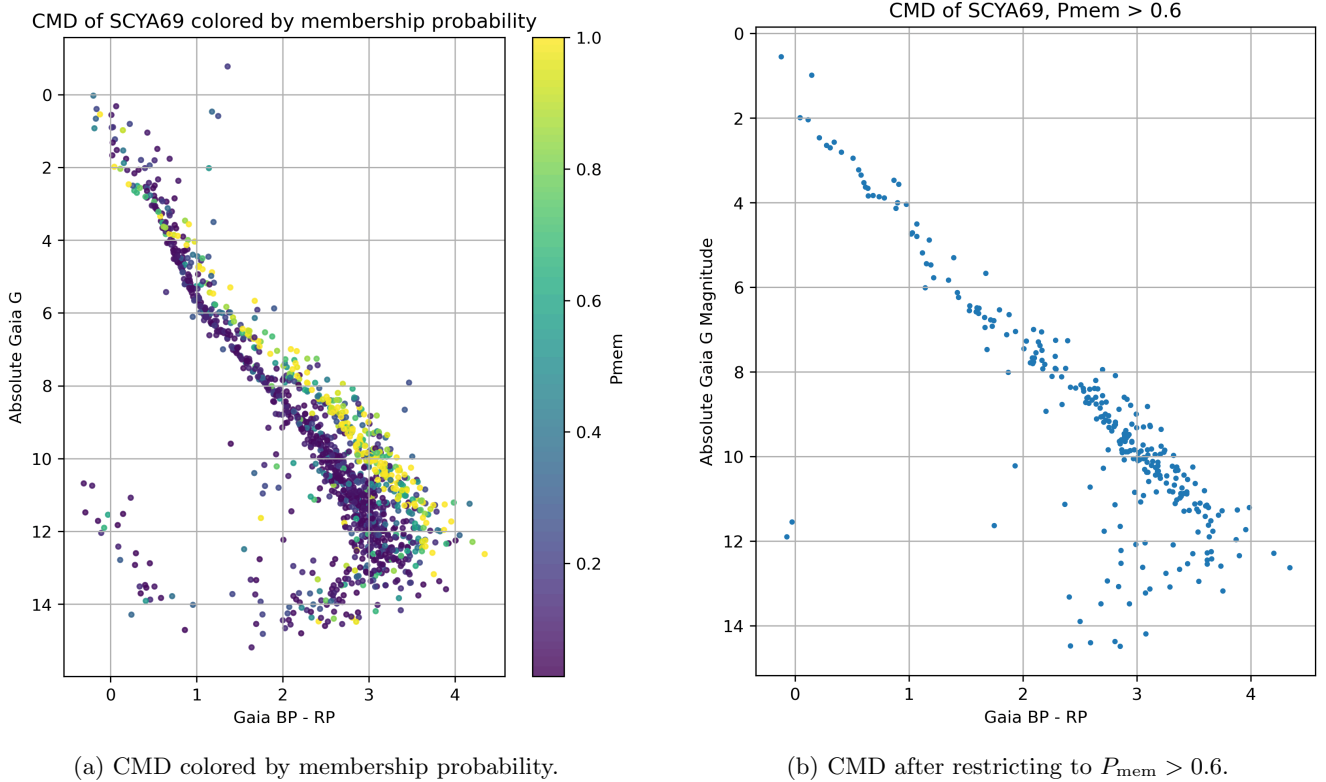


Figure 4: Color-magnitude diagrams of SCYA69 using Gaia absolute G magnitude and Gaia BP-RP color.

(F) I investigated how the three membership markers (Pyoung, strength, and founding) are distributed in three different spaces: sky position, proper motion, and the color-magnitude diagram (CMD). My goal was not only to compare the markers directly, but also to look for evidence that the selected stars form a physically meaningful young population.

We created two plots, both using a controlled cut on strength to restrict the sample to stars above a chosen threshold (for example, strength > 0.5). This restriction was motivated by the SPYGLASS IV approach, where increasing the strength threshold helps isolate the central population of the group and reduce field contamination, allowing the young sequence to emerge more clearly.

The first plot was a CMD showing the restricted sample after applying the strength cut. By examining the CMD under progressively stronger cuts, we could test whether a clearer main sequence or pre-main-sequence population began to emerge. We additionally overplotted the stars selected by strength (red) and the stars flagged as founding (yellow) to check whether these stars occupied CMD positions consistent with a young stellar population.

The second plot was a 3-by-3 comparison figure. The rows represented three different spaces: spatial position, proper motion, and CMD position. The columns represented the three membership markers: Pyoung, strength, and founding. For Pyoung and strength, we used continuous colour scales to show how the value of each marker varied across the data. Since founding is already a binary flag, founding and non-founding stars were plotted separately.

This section involves investigating how the distribution of all three markers differ and if they had evidence of being a real young cluster for the population SCYA 69:

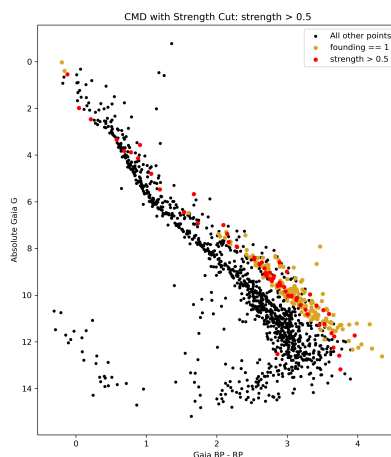


Figure 5: CMD of SCYA69 with strength cut of 0.6 applied. Red dots represent filtered data points, yellow foundation points, and black field data

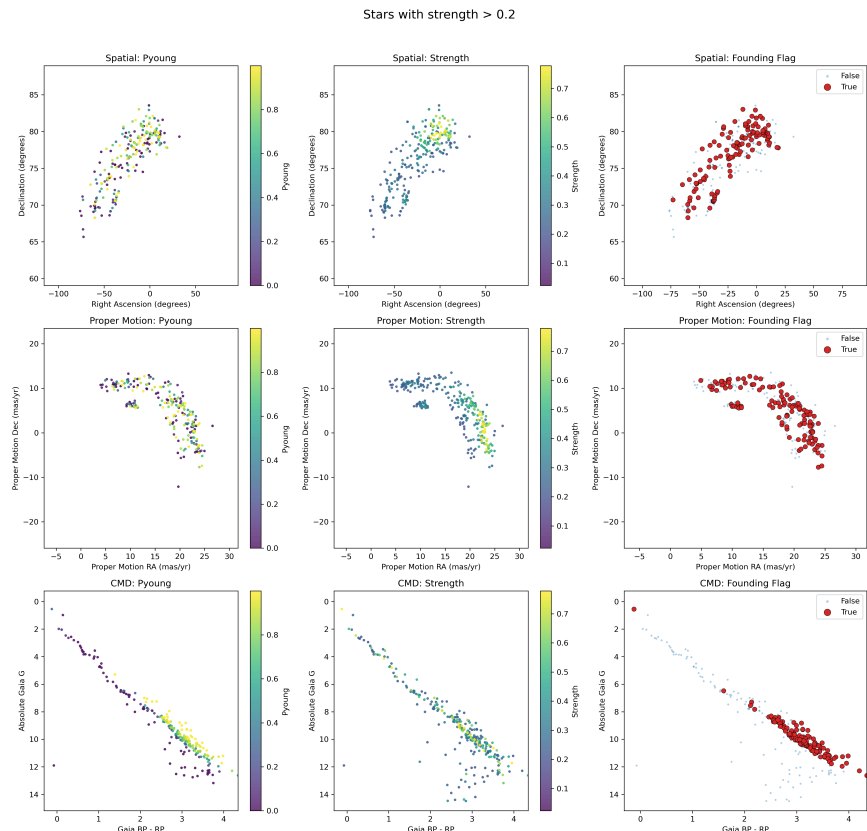


Figure 6: 3x3 plot of SCYA69 for membership markers Pyoung, Strength, Foundation with strength cut of 0.2 applied

From figure 5 applying a strength cut of 0.6, we see that restricting in strength reveals a smooth sequence consistent with the pre-main sequence branch. Importantly, the founding stars identified by HDBSCAN were also pre-main sequence stars and O/B stars, giving strong evidence to this cluster being a young one.

Figure 6 gives us a 3x3 sub figure when applying a strength cut of 0.2.

Spatial Distribution:

Pyoung stars seem to be relatively spread dispersed in spatial and proper motion distribution as compared to strength. Strength provides much more clustering as expected since it measures proximity with respect to the founding stars. When identifying a genuine young population, it is important that the region of highest strength overlaps with an overdensity of young stars in the Pyoung distribution. Although the young stars are more extended overall, this may simply indicate that the surrounding region contains many young stars. The key result is that the area of highest strength coincides with a concentration of high Pyoung stars, providing evidence for the presence of a real young cluster.

Proper Motion:

The same idea applies with proper motion. A genuine young cluster should exhibit coherent proper motions because its stars formed from the same molecular cloud and therefore share a common kinematic motion. We observe that the region with the highest strength values also corresponds to the concentration of high Pyoung stars in proper motion space. This agreement suggests that the cluster is not only spatially concentrated, but also kinematically coherent.

CMD:

Pyoung stars seems like the strongest discriminator in color-magnitude space. We see a steady rise from stars on the main sequence to the pre main sequence, with their probability of being young consistently increasing indicating that they are correctly distinguishing young from old.

The strength CMD seems to be more broadly selected and less confined to the PMS branch. We see that there are some high strength candidates still selected higher along the CMD because they are deeply embedded in the cluster even though not young. Since most stars embedded in the cluster will be young (as we can see), there still may be some that are older and so they are still being selected as we move up the main sequence

The founding flag identifies many stars along the pre-main-sequence branch while also recovering O/B-type stars. Since the founding population is designed to incorporate spatial, kinematic, and photometric information simultaneously, this produces a promising distribution that is broadly consistent with a genuine young stellar population.

- (G) We will now perform this analysis on the other three populations CG-26 (S-Sco), SCYA6 (Circinus) and SCYA72:

0.0.1 CG-26

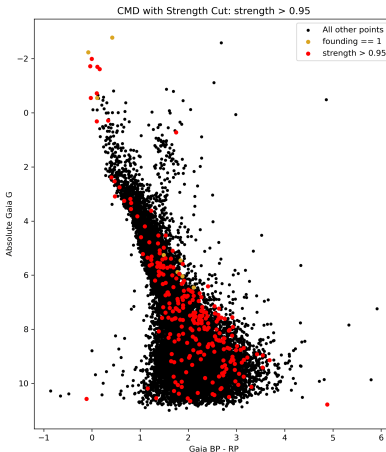


Figure 7: CMD of CG-26 with strength cut of 0.95 applied.

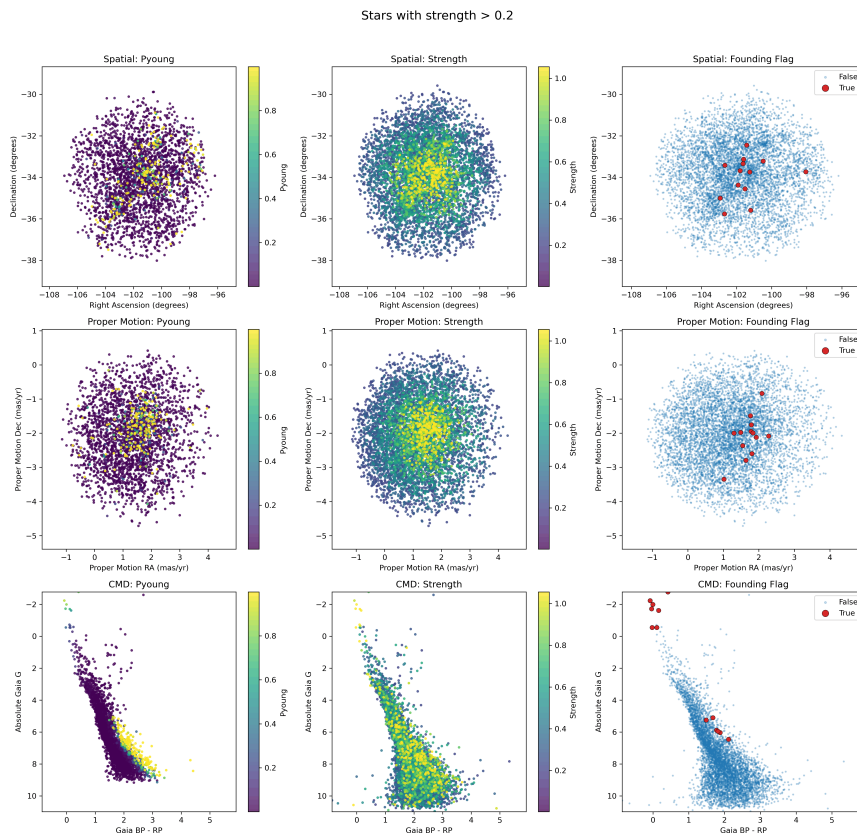


Figure 8: 3x3 plot of CG-26 for membership markers Pyoung, Strength, Foundation with strength cut of 0.2 applied

Looking at Figure 7, when we restrict in strength on the cluster, we expect to get a more clear, smooth pre main sequence. However, here even when restricting strength to 0.95, the filtered stars still create a structure that resembles that of the field sequence.

Furthermore, the foundation stars identified impose some concern as well. The foundation flag was able to identify a few O/B stars, which is normally a strong sign of a young cluster. However, they also identified just a few stars sitting in the middle of the CMD. We would expect more stars to be identified in the lower right with the pre-main sequence (PMS), but this identification doesn't happen. The lack of young structure in the strength restriction paired with the foundation stars makes me guess that the cluster was possibly identified by reddening anomalies.

For the 3x3 plot, I did not impose a super large strength cut on this 3x3 plot so I could better analyze the features of a larger cluster.

We see similar distributions as in the SCYA69 population:

Since strength measures how centrally concentrated a star is relative to the founding population and therefore naturally emphasizes the cluster core. The Pyoung stars show a surprising correlation of density in proper motion and in spatial distribution corresponding to the cluster core, which is a positive indicator that the cluster may be young.

The strength distribution in the CMD is broader and less confined to the PMS. Some high-strength stars remain positioned higher along the main sequence because they are deeply embedded within the cluster environment despite not necessarily being young themselves. Pyoung provides a stronger discrimination in color-magnitude space. The CMD shows a gradual increase in youth probability moving from the main sequence toward the pre-main-sequence branch.

0.0.2 SCYA6

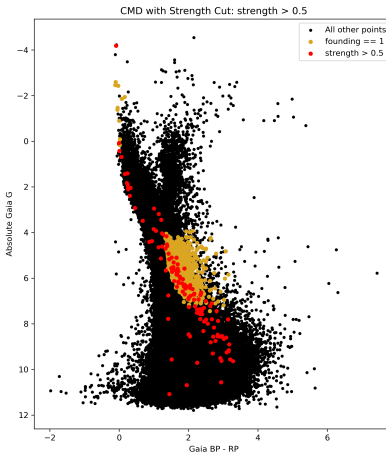


Figure 9: CMD of SCYA6 with strength cut of 0.7 applied.

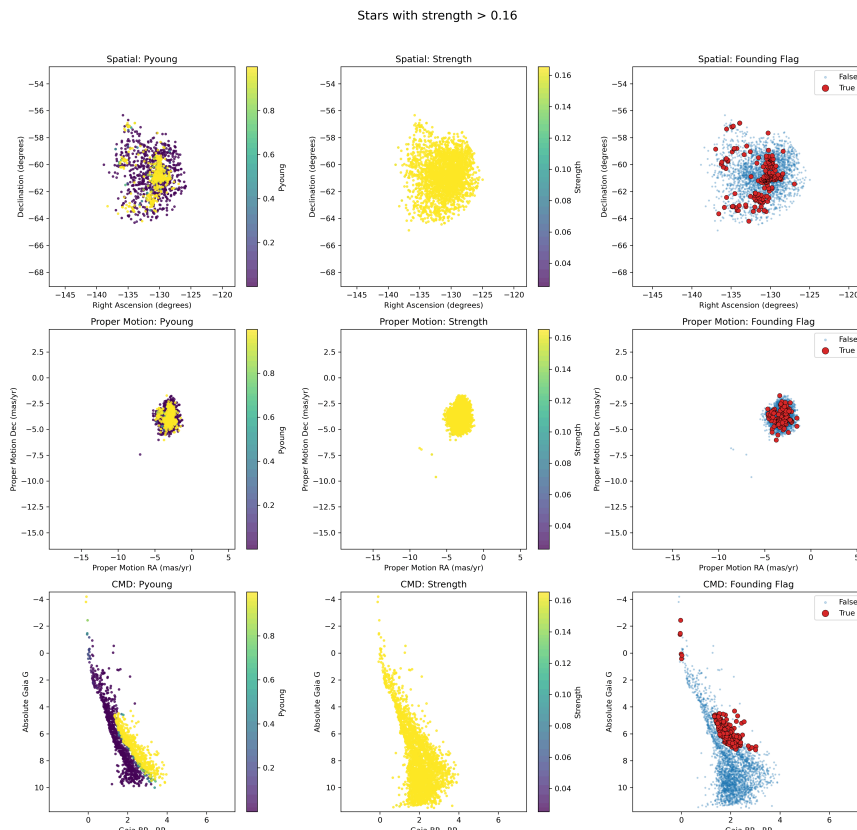


Figure 10: 3x3 plot of SCYA6 for membership markers Pyoung, Strength, Foundation with strength cut of 0.16 applied

In Figure 9, applying a restriction in strength reveals a much more clearly defined pre-main-sequence population embedded within the larger field population. The sequence becomes significantly easier to distinguish once lower-strength stars are removed, suggesting that the strength parameter is effective at isolating the central young population of the group.

The CMD also shows that many of the founding stars cluster down and to the right of the subgiant branch. This distribution requires further investigation, since we would generally expect a larger fraction of founding stars to appear along the lower-mass pre-main sequence. Their location raises the possibility that reddened older stars are being shifted into regions of the CMD that mimic young stellar populations. However, the presence of several OB founding stars is a strong positive indicator, since massive OB stars are reliable tracers of youth and are unlikely to belong to an old field population.

In Figure 10, when we restrict in strength, we see a relatively large field population restrict itself nicely in spatial and proper motion space. Spatially, when we restrict to the center of the cluster there is a strong concentration of stars with high Pyoung values. Additionally, with this imposed restriction in proper motion space there is an even tighter cluster of such young stars with similar proper motions, which is what is expected of a young population.

With no restriction in strength, the 3x3 gets much more confusing. Through further analysis and discussion with Dr. Kerr, we identify that this cluster is in fact real, but embedded within a highly reddened cluster of stars.

It appears as if there is this weird filamentary structure of young stars the don't quite look at though they correlate to a young cluster. A large proportion of these stars were identified by the founding flag, and was thus why they showed up on the CMD in a concerning spot where we know reddening may have occurred right under the sub-giant branch where old stars may get pushed to looking young. Since when we restrict in

strength, these stars go away, we know that they are not actually located in the center of the cluster, but are distributed in other areas not consistent with a truly young cluster. The true young population appears once we restrict in strength.

0.0.3 SCYA72

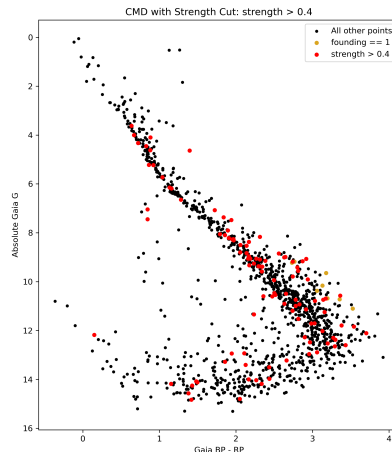


Figure 11: CMD of SCYA72 with strength cut of 0.95 applied.

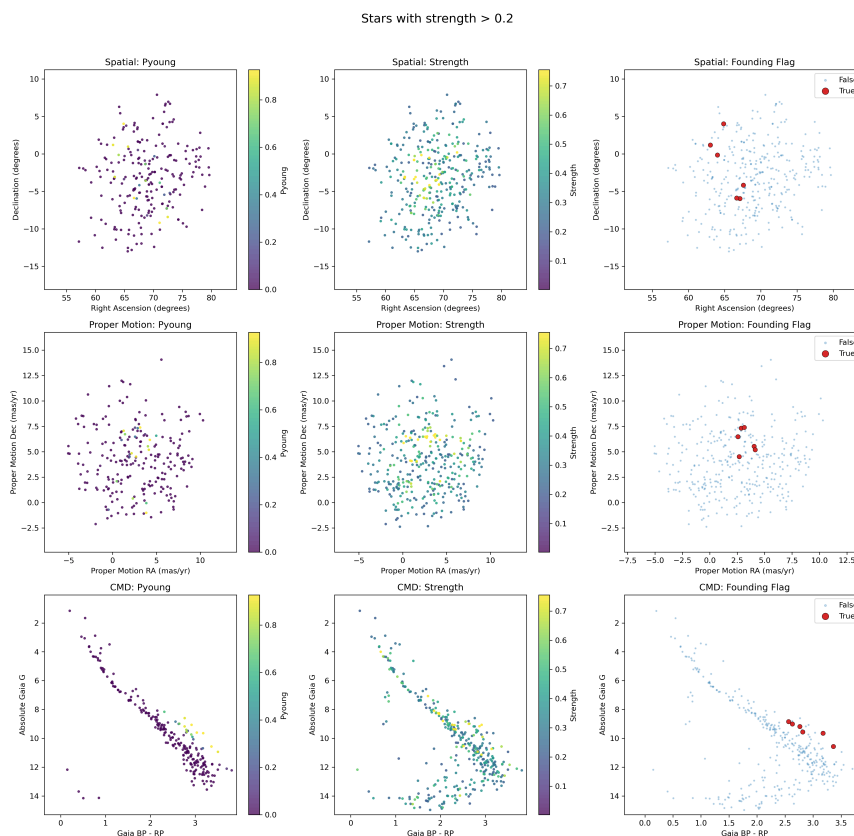


Figure 12: 3x3 plot of SCYA72 for membership markers Pyoung, Strength, Foundation with strength cut of 0.2 applied

When restricting in strength, the CMD of SCYA72 shown in Figure 11 does reveal a pre-main sequence, but it is not isolated. There also exists a distinct field distribution of stars, and no further amount of restriction makes the pre-main sequence more prominent and the field less prominent.

The founding flags do identify a few stars and all of them sit on the pre-main sequence which is a positive sign.

These conflicting signals may be alleviated through analysis of the 3x3 plot in Figure 12.

When restricting the 3x3 plot to a strength ≥ 0.2 , there doesn't seem to be a clustering of young stars close anywhere, and no clustering especially close to where the strength is highest in both spatial and proper motion space. Founding stars are not especially clustered spatially, however do have relatively similar proper motions.

Since the founding stars are the original high-confidence seed population used to identify the group, seeing them align with the pre-main sequence is a strong positive sign. It seems that the candidate may represent a real but diffuse young population whose separation from the field is comparatively weak.

1 Part 2

(A) Done.

(B) First set of downloaded isochrones:

In $\log(\text{age}/\text{yr})$:

initial value: 6.0 final value: 7.9 step: 0.1

Producing a set of isochrones from 1Myr to 80 Myr

Second set:

In $\log(\text{age}/\text{yr})$:

initial: 9.0 final: 10.1 step: 0.2

Producing a set of isochrones greater than 1 Gyr in age.

(C) Created a new column for Age in Myr by converting the log Age in Age

(D) Plotting these isochrones over CMD's from Part 1 of the assignment. For now doing this from an eye by eye comparison.

In future implementation, it would be best to find a least squares fit to fit an isochrone to the pre-main sequence and an isochrone to the field distribution.

For the pre-main sequence we do a least-squares fit of the restricted population and for the field we do a similar least-squares fit of a specific subset that have low Pmem, were not identified as founding stars, and strength restricted stars.

General Systematic Deviations Seen:

The main deviation occurs in the bottom-left tail of the CMD. There exists a population of faint stars around approximately $\text{BP}-\text{RP} \sim 0-1$ and $M_G \sim 11-14$. These stars do not follow either the young pre-main-sequence isochrone or the older field main-sequence population. They appear too blue and faint compared to the expected stellar sequences, suggesting that they are unlikely to belong to the young association itself. It is also unlikely that these stars simply represent an older stellar population shifted into this region through normal stellar evolution. More plausibly, they are caused by some combination of field contamination, poor photometry, distance or parallax uncertainties, unresolved binaries, or other observational effects.

Another deviation is that the PMS is not a thin line. Ideally, with perfect measurements for a population of a certain age we expect a thin isochronal line given by the IMF. For many reasons such as unresolved binaries, photometric and astrometric uncertainties, extinction broadening etc... the data forms a spread about the young isochrone, especially at redder colors (harder to measure well).

For the eye by eye comparison, see some examples below:

1.1 SCYA69

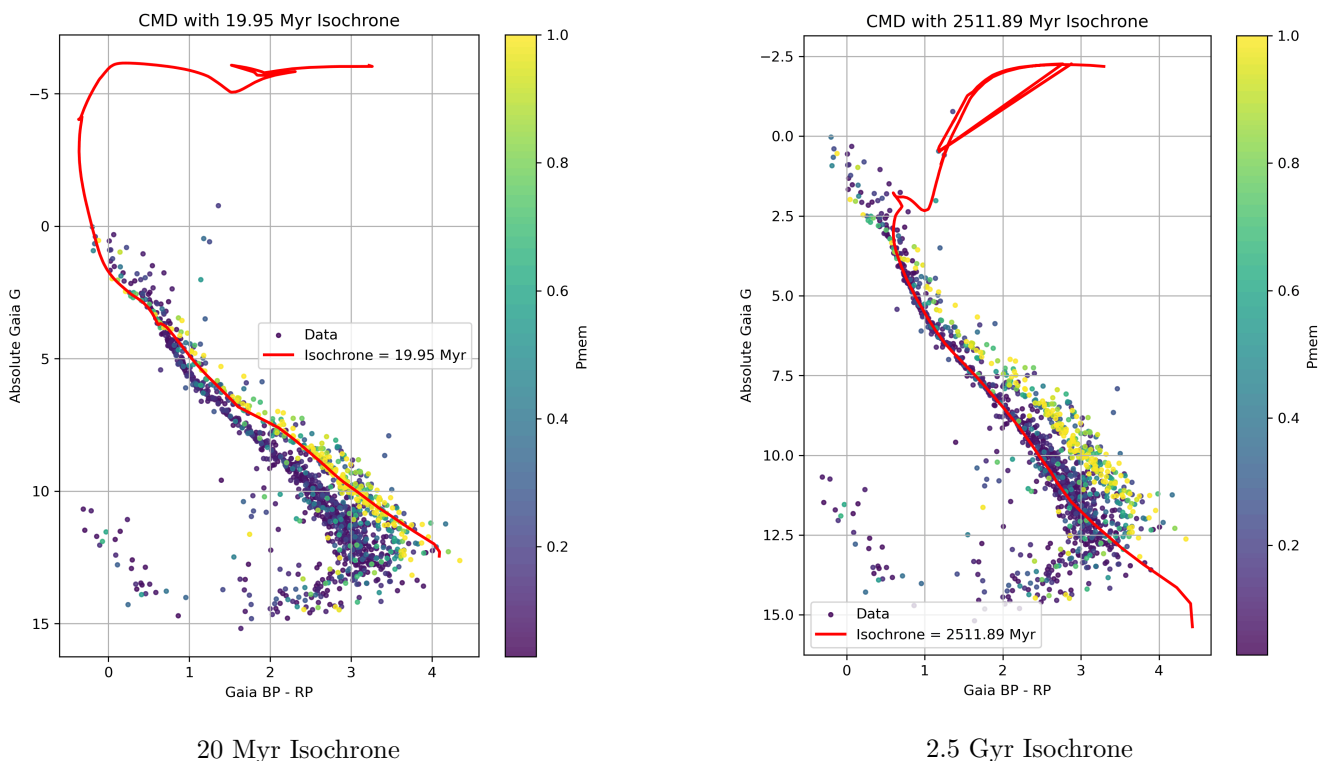


Figure 13: CMD of SCYA69 with 20 Myr and 2.5 Gyr isochrones overlaid.

PMS:

31.62 Myr isochrone seems to fit best because it follows both the upper-left main sequence and the lower-right PMS population simultaneously. In the upper left where the stars are brighter and bluer with a high probability of membership, the isochrone passes through the sparse points. Towards the lower-right PMS region the isochrone traces along the lower middle edge of the dense high Pmem PMS population rather than lying significantly above or below it.

Field:

The field distribution is best fit by the 1 Gyr isochrone, although it may be slightly younger since 1 Gyr is the youngest old isochrone included in the downloaded set. The 1 Gyr isochrone follows the center of the main field-star sequence reasonably well across much of the CMD, especially in the region around BP–RP ~ 2 –3.5. The older 2.5 Gyr isochrone begins to deviate toward redder and fainter values, suggesting that the dominant field population is better represented by an age closer to 1 Gyr.

1.2 SCYA6

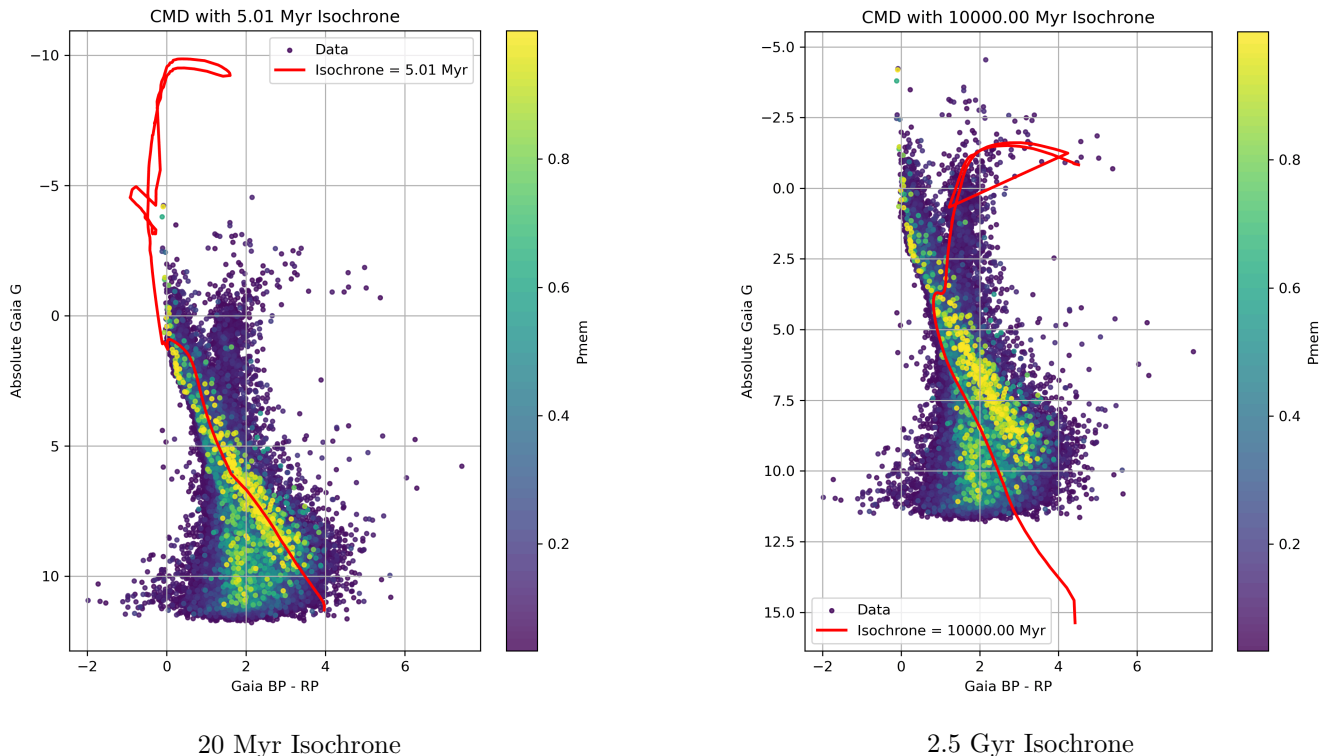


Figure 14: CMD of SCYA6 with 20 Myr and 2.5 Gyr isochrones overlaid.

PMS:

The best eye to eye fit of the pre-main sequence occurs at 5Myr where it is able to fit to the center of the width of the distribution in the lower right arm as well as follow the fit as it moves up the high Pmem members.

Field:

It was much harder to fit an isochrone for the field population. Since the distribution is much wider, and we haven't imposed any cuts as mentioned above, there was nothing that would perfectly match the field sequence. Fitting an isochrone would be much better with an implementation of a least-squares fitting model.

- (E) Initially, I was having difficulties fitting the SCYA datasets with the Rottensteiner empirical isochrones because many of the isochrones did not extend far enough into the faint and red regions of the CMD. To investigate this more carefully, I re-evaluated the CMD coverage of the Rottensteiner empirical isochrones using a more realistic coverage criterion rather than requiring every extreme outlier in the data to be reproduced exactly.

The comparison was restricted to the young Rottensteiner isochrones, defined here as populations with ages ≤ 80 Myr, since these are the relevant empirical sequences for young SCYA-like populations. This produced a sample of 59 young empirical isochrones. The remaining 34 Rottensteiner isochrones correspond to populations older than 80 Myr and are therefore more representative of older field populations rather than young stellar associations.

To determine whether an isochrone provided sufficient CMD coverage, I compared its span in Gaia BP–RP colour and absolute Gaia G magnitude against the observed SCYA member locus. Where available, I restricted the data to stars with

$$P_{\text{mem}} > 0.5$$

to reduce field contamination. Instead of using the absolute maximum values of the CMD distribution which gave me problematic results, I used the 99th percentile of the data to create some buffer of the low mass points.

Using this buffered criterion, 17 out of the 59 young Rottensteiner isochrones were found to have sufficient CMD coverage for the SCYA datasets. These are:

Acrux, rho_Lup, eta_Lup, nu_Cen, sig_Cen, phi_Lup, US-foreground, IC_2391, IC_2391_C2, Platais_8, NGC_2451A_C2, IC_2602, IC_2602_C2, Platais_9_C2, Melotte_20_C2, Melotte_22, Melotte_22_C2

This range covers the main SCYA CMD sequence well, including much of the important low-mass and pre-main-sequence region. Although it does not reach the absolute faintest edge of every dataset, particularly the lowest-mass tail of SCYA69, the overall overlap is sufficient for it to remain useful for fitting and generating young stellar populations.

Overall, a subset of 17 young isochrones do provide adequate CMD coverage when a reasonable buffer criterion is applied. These isochrones are therefore the strongest candidates for comparison and population generation, while the older Rottensteiner isochrones should generally be excluded because they are more representative of older field-like populations rather than young SCYA-like stellar associations.

See an example below of a Rottenstein Isochrone fitting the pre-main sequence and an older isochrone fitting the field:

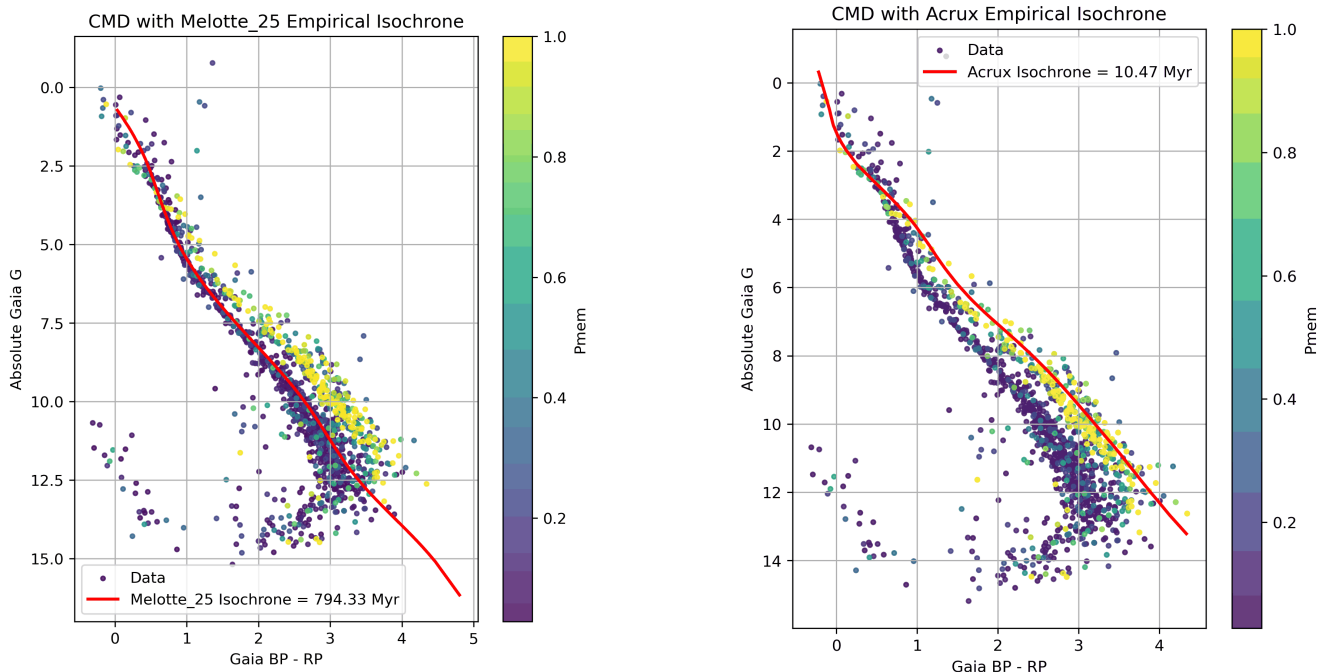


Figure 15: CMD of SCYA69 with a) Acrux 10.47 Rottenstein empirical isochrone overlaid and b) Melotte-25 794.33 Myr empirical isochrone overlaid

(F) Through the analysis performed in Section E, we determined that 17 out of the 59 young Rottensteiner isochrones provide sufficient coverage of the pre-main-sequence population.

For the field sequence, there were 34 older isochrones identified. For SCYA69 and SCYA72, which extend significantly further down the CMD than SCYA6 and CG26, only two old isochrones — **Melotte.111** and **Melotte.25** — extend far enough to adequately overlay the field sequence. In contrast, for SCYA6 and CG26, there are approximately 5 and 10 old isochrones respectively that are sufficiently extended to represent the field population.

References

CMD 3.9 Input Form, stev.oapd.inaf.it/cgi-bin/cmd. Accessed 19 May 2026.

Kerr, Ronan M., et al. “Stars with photometrically young gaia luminosities around the solar system (spyglass). I. Mapping Young Stellar Structures and their star formation histories.” *The Astrophysical Journal*, vol. 917, no. 1, 1 Aug. 2021, p. 23, <https://doi.org/10.3847/1538-4357/ac0251>.

Kerr, Ronan, et al. “Spyglass. iv. New Stellar Survey of recent star formation within 1 KPC.” *The Astrophysical Journal*, vol. 954, no. 2, 29 Aug. 2023, p. 134, <https://doi.org/10.3847/1538-4357/ace5b3>.

Rottensteiner, Alena, and Stefan Meingast. “An empirical isochrone archive for nearby open clusters.” *Astronomy and Astrophysics*, vol. 690, 27 Sept. 2024, <https://doi.org/10.1051/0004-6361/202347701>.